

Algebra II with Trigonometry

Chapter 6.5

Olympiad Problems (Gemini)

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[4533652_alg-2trig-h-chp-6-5-note.pdf](#)

Problems

1. **Phase Shift Estimation:** Evaluate exactly:
 $\arcsin(\sin 10) + \arccos(\cos 10)$. (Assume the angle 10 is in radians).
2. **Nested Invariant:** Find the exact value of $\tan(\arcsin(\cos(\arctan 2024)))$.
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3. **Algebraic Optimization:** Find the maximum possible value of the function $f(x) = (\arcsin x)^3 + (\arccos x)^3$ over its real domain.
4. **Trigonometric Equation:** Determine the unique positive real number x that satisfies the equation $\arcsin(x) + \arcsin(2x) = \arccos(x)$.
5. **Telescoping Arguments:** Evaluate exactly, using purely mental arithmetic: $\arctan\left(\frac{1}{3}\right) + \arctan\left(\frac{1}{5}\right) + \arctan\left(\frac{1}{7}\right) + \arctan\left(\frac{1}{8}\right)$.
6. **Periodic Cancellation Graphs:** Determine the total combined length of all intervals within the domain $x \in [0, 100\pi]$ for which the equation $\arcsin(\sin x) = \arccos(\cos x)$ holds true.